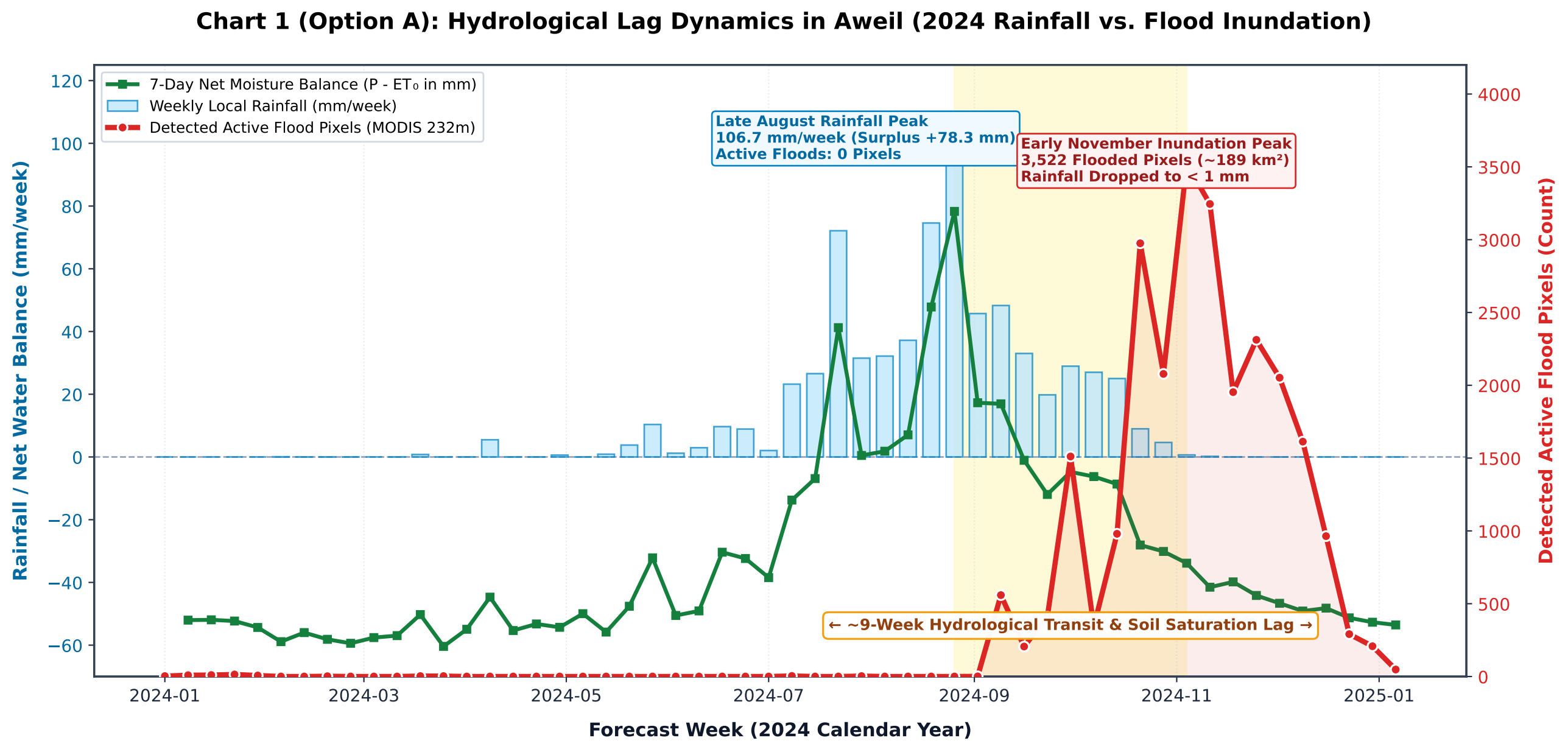
Chart 1 (Option A): Hydrological Lag Dynamics in Aweil (2024 Rainfall vs. Flood Inundation)
7-Day Net Moisture Balance (P - ET₀ in mm)
Weekly Local Rainfall (mm/week)
Detected Active Flood Pixels (MODIS 232m)
Late August Rainfall Peak
106.7 mm/week (Surplus +78.3 mm)
Active Floods: 0 Pixels
Early November Inundation Peak
3,522 Flooded Pixels (~189 km²)
Rainfall Dropped to < 1 mm
← ~9-Week Hydrological Transit & Soil Saturation Lag →
Rainfall / Net Water Balance (mm/week)
Detected Active Flood Pixels (Count)
Forecast Week (2024 Calendar Year)
120
100
80
60
40
20
0
−20
−40
−60
4000
3500
3000
2500
2000
1500
1000
500
0
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
2025-01